Felipe Portugal

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Portfolio: felipe-portugal.github.io/obsidian-glass

Profile: Software Engineering student at ÉTS with a college diploma (DEC) in computer science technology. Internship and then employment in web and mobile development at ITMI, plus robotics experience with Boston Dynamics' Spot robot. Interested in embedded systems and applied AI.

Availability: Available for a 4- or 8-month full-time internship starting January 2027; also open to full-time positions.

Education

Bachelor of Engineering, Software Engineering
École de technologie supérieure (ÉTS), Montreal | since 2024 | expected graduation: May 2029

Diploma of College Studies (DEC), Computer Science Technology
Cégep de Sept-Îles | graduated in 2024

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C#, PHP, SQL, Rust, Bash
Web and Mobile: React Native, Expo, ASP.NET Core (MVC, Blazor, Web API), REST APIs, HTML, CSS, Android, JavaFX
Data: PostgreSQL, SQL Server, MySQL, SQLite, InfluxDB, Grafana, data modelling
Quality and Tools: pytest, JaCoCo, TDD, Git, Gitea, GitLab, Azure DevOps Boards, Docker, Linux, Agile/Scrum
Robotics, Embedded and AI: Spot SDK, PyTorch, OCR, OpenCV, NumPy, Raspberry Pi, Arduino, MQTT, Bluetooth LE, LoRaWAN

Professional Experience

Institut technologique de maintenance industrielle (ITMI) — *applied research centre in industrial maintenance*
Sept-Îles, Quebec | January 2025 – September 2025

Assistant Developer, Industrial Data Visualization | *May 2025 – September 2025*
Developer Intern, Industrial Data Visualization | *January 2025 – May 2025*

Shared contributions across both roles — January to September 2025

- Contributed to the development of the mobile application of a parking management system deployed for all Cégep de Sept-Îles staff, using React Native, Expo and TypeScript.
- Developed pages of the same system's administration web interface, as well as features of a web application for inventory and contract management.
- Designed and implemented a PostgreSQL module for Nomad, ITMI's internal framework, mainly in PHP; contributed to its REST APIs and backend components (queries, migrations) in Docker-based projects.
- Designed SQL Server schemas and queries; developed industrial data visualizations with PHP and Grafana.
- Worked in Agile/Scrum with a TDD approach: pull requests, code reviews and branches in Gitea; task tracking in Azure DevOps Boards.
- Maintained applications in production, trained users and worked with clients.

Technical Advisor — Embedded Systems
Summer 2024 | ITMI, Sept-Îles, Quebec

- Developed a working Python prototype with the Spot SDK to select existing autonomous missions of the Spot robot and record new ones.
- Automated the processing of mission photos on a Linux server: gauge detection with PyTorch, OCR reading of their values and image processing with OpenCV, Pillow and NumPy; exported the readings to Excel and CSV files accessible to staff.
- Collected outdoor sensor data in Rust, stored it in InfluxDB and visualized it in Grafana.
- Wrote automated client and server tests in Python with pytest, including load tests; wrote technical documentation and trained staff.

Research Assistant
Winter 2024 | ITMI, Sept-Îles, Quebec

- Built a Python proof of concept with the Spot SDK, with testing and validation using sensors and payloads; modelled the explored environment from the robot's lidar point clouds.

Computer Operator
Summer 2023 (internship), Fall 2023 (employment) | CISSS de la Côte-Nord, regional health and social services centre, Sept-Îles, Quebec

- Replaced and configured workstations; provided hardware and software support; deployed Windows updates.

Selected Projects

- **Software Testing and Quality — Java Application** *(2025, team, LOG240, ÉTS)*: change impact analysis; black-box, unit and integration testing; 99% coverage of the AccountDAO and CategoryDAO classes (JaCoCo); defect tracking in GitLab.
- **JavaFX Image Manipulation Application** *(2025, team, LOG121, ÉTS)*: zoom, translation, undo and save; MVC architecture with three views and the Observer pattern.
- **Connected Spot Robotics Platform** *(2023, team of two, DEC)*: recorded Autowalk mission, with the robot oriented toward the heat source spotted by the thermal camera; collection of audio, images and sensor data; Docker container launched over SSH on the robot and a Bash script.
- **Android Monitoring System** *(2024, individual, DEC)*: Android application in Java, Raspberry Pi in Python and Arduino; requirements document and deliverable validation with the client.
- **ASP.NET Core Web Applications** *(2023–2024, DEC)*: MVC applications with Entity Framework and SQL Server; REST Web API consumed by a Blazor client and deployed on IIS.
- **Web Sites with Authentication** *(2024, DEC)*: PHP and MySQL: sessions, roles, PDO prepared statements, OAuth (Google, Facebook, GitHub) and QR-code two-factor authentication.
- **Internet of Things** *(2023–2024, DEC)*: motion detection with Android, Raspberry Pi and MQTT (SQLite history); Arduino labs with Bluetooth LE and LoRaWAN.

Involvement

- **CAPRA — Software Team** *(Fall 2024, student club, ÉTS)*: research on components for robotic platforms; hardware performance and compatibility testing.
- **SLAN Révolution 2024 — LAN Party** *(2023–2024, organizing team, Cégep de Sept-Îles)*: 24-hour event with more than 200 participants; designed and set up the electrical and network plans, managed participant and equipment logistics, sought sponsors and provided technical support during the event.

Languages

French, English and Spanish.